

(Probability) Generating functions for complex networks

Reference:
Chapter 12, Networks, by Mark Newman

Motivation and definition

Motivation: we have already been using it (i.e. it is useful!).

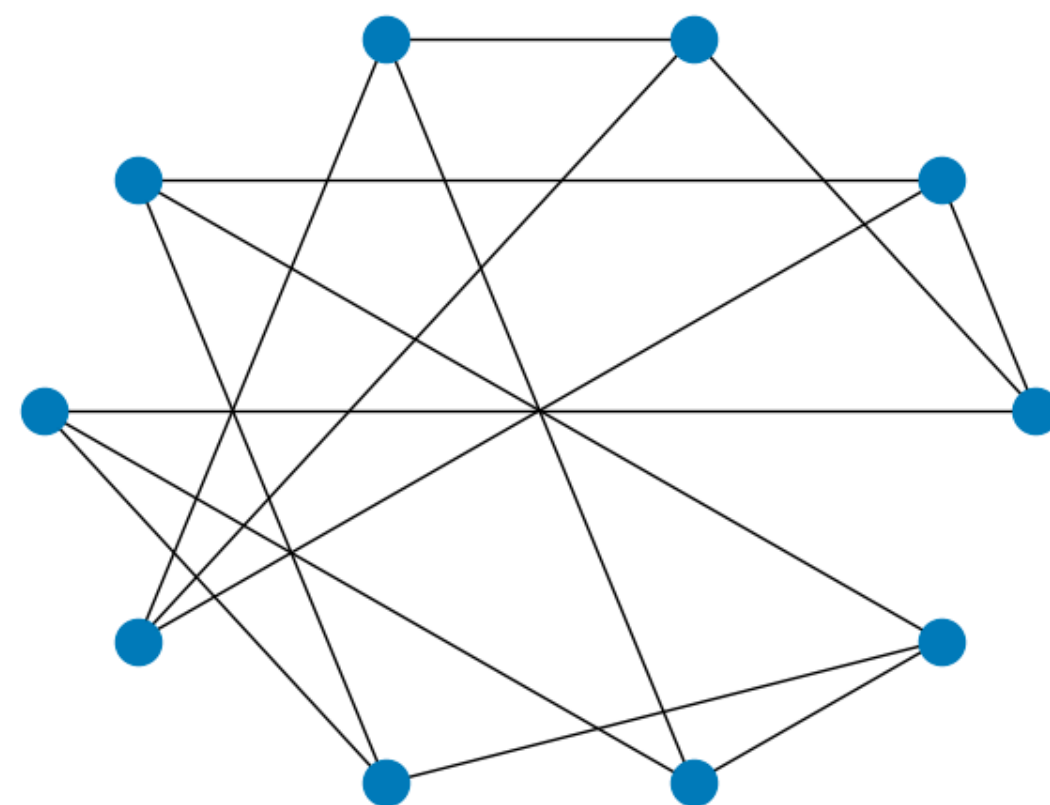
Probability distribution of degrees in a network: p_k

Random approximation: p_k fully describes the network structure.

Degree probabilities p_k in a network can be used to create a polynomial function:

$$g(z) = p_0 + p_1 z + p_2 z^2 + p_3 z^3 + \dots = \sum_{k=0}^{\infty} p_k z^k \xrightarrow{\text{To generate the probabilities}} p_k = \frac{1}{k!} \left. \frac{d^k g}{dz^k} \right|_{z=0}$$

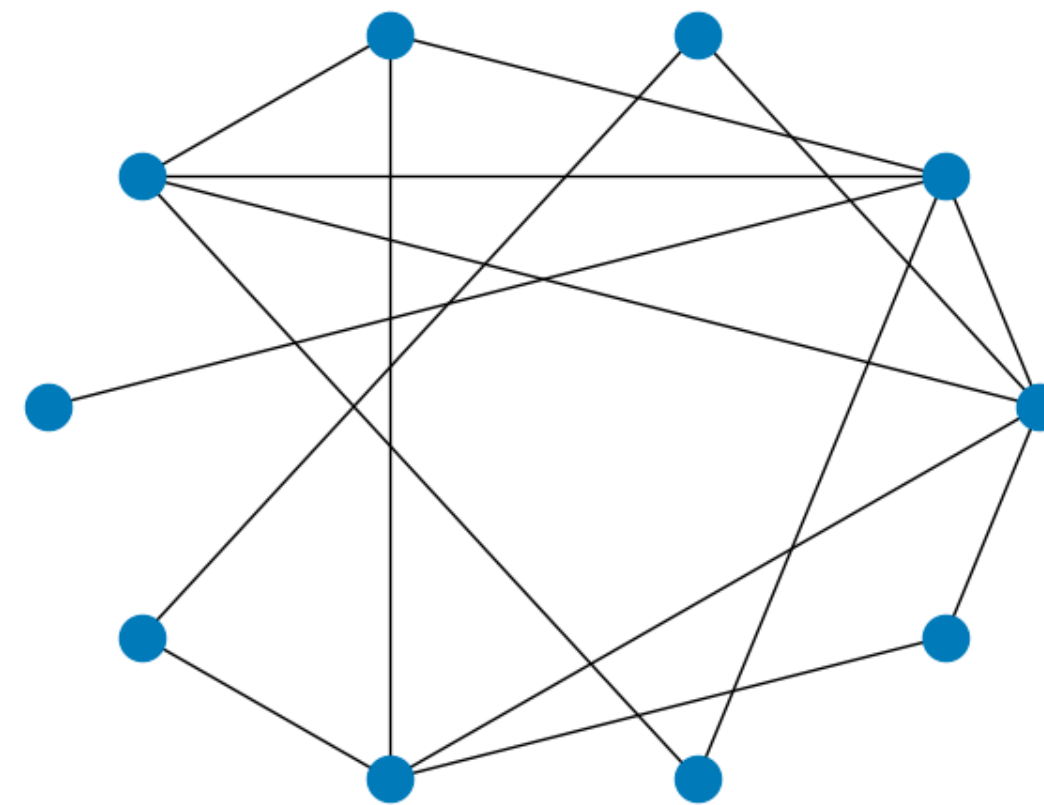
Example 1: Random regular network:



$$p_k = \begin{cases} 1, & \text{if } k = 3 \\ 0, & \text{otherwise} \end{cases}$$

$$g(z) = z^3$$

Example 2: Erdos Renyi network with average degree c :



$$p_k = e^{-c} \frac{c^k}{k!}$$

$$g(z) = \sum_{k=0}^{\infty} e^{-c} \frac{c^k}{k!} z^k = e^{-c} \sum_{k=0}^{\infty} \frac{(cz)^k}{k!} = e^{c(z-1)}$$

Normalisation and moments (probabilities)

$$g(z) = \sum_k p_k z^k$$

0th moment: total probability must sum up to 1:

$$g(1) = \sum_{k=0}^{\infty} p_k \longrightarrow \boxed{g(1) = 1}$$

1st moment: average degree

$$g'(z) = \sum_{k=0}^{\infty} k p_k z^{k-1} \longrightarrow \boxed{g'(1) = \sum_{k=0}^{\infty} k p_k = \langle k \rangle}$$

2nd moment

$$z \frac{d}{dz} \left(z \frac{dg}{dz} \right) = \sum_{k=0}^{\infty} k^2 p_k z^k$$



$$\boxed{\langle k^2 \rangle = \left[\left(z \frac{d}{dz} \right)^2 g(z) \right]_{z=1}}$$

***m*th moment**

$$\boxed{\langle k^m \rangle = \left[\left(z \frac{d}{dz} \right)^m g(z) \right]_{z=1}}$$

Products of generating functions

Given m integers $k_1, k_2, \dots, k_m$, each with independent random distribution $p_k^{(1)}, p_k^{(2)}, \dots, p_k^{(m)}$

The generating function of their sum is simply the product of their generating functions.

Probability that sum is equal to s :

$$\pi_s = \sum_{k_1=0}^{\infty} \dots \sum_{k_m=0}^{\infty} \delta(s, \sum_i k_i) \prod_{i=1}^m p_{k_i}^{(i)}$$

$$\delta(s, \sum_i k_i)$$

Kronecker delta to ensure $s = \sum_i k_i$

Generating function of probabilities π_s :

$$h(z) = \sum_{s=0}^{\infty} \pi_s z^s = \sum_{s=0}^{\infty} z^s \sum_{k_1=0}^{\infty} \dots \sum_{k_m=0}^{\infty} \delta(s, \sum_i k_i) \prod_{i=1}^m p_{k_i}^{(i)}$$

$$h(z) = \sum_{k_1=0}^{\infty} \dots \sum_{k_m=0}^{\infty} z^{\sum_i k_i} \prod_{i=1}^m p_{k_i}^{(i)} = \prod_{i=1}^m \sum_{k_i=0}^{\infty} p_{k_i}^{(i)} z^{k_i}$$

$$\downarrow$$

$$= \prod_{i=1}^m g^{(i)}(z),$$

$$g^{(i)}(z) = \sum_{k=0}^{\infty} p_k^{(i)} z^k$$

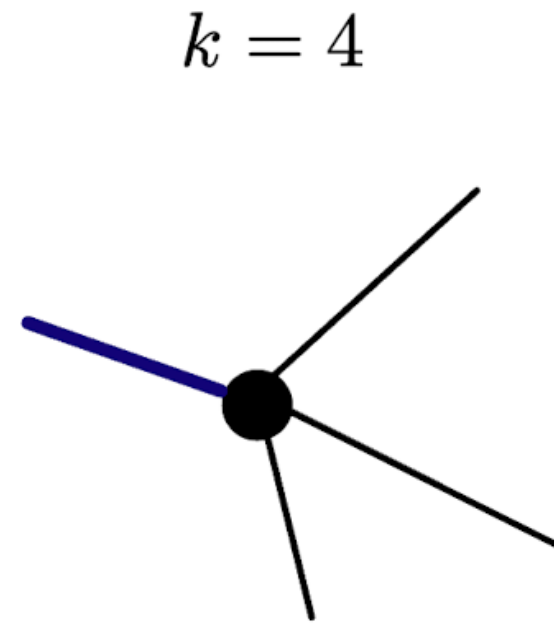
Generating function for sum of m i.i.d distributions:

$$h(z) = [g(z)]^m$$

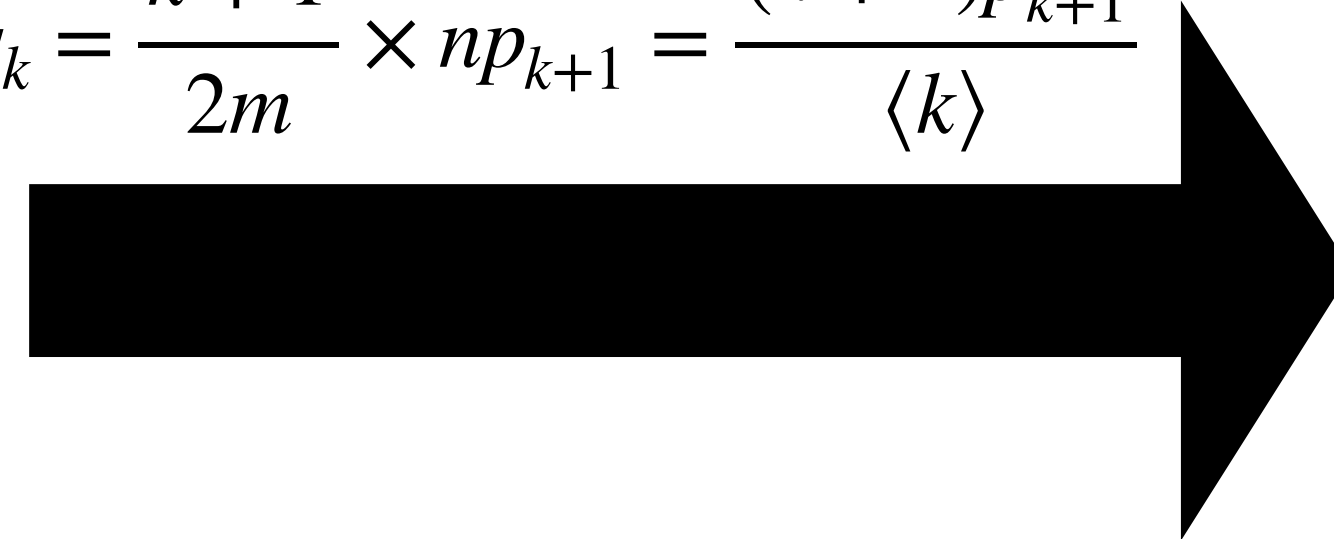
In the context of networks

Degree distribution:

$$g_0(z) = \sum_{k=0}^{\infty} p_k z^k$$



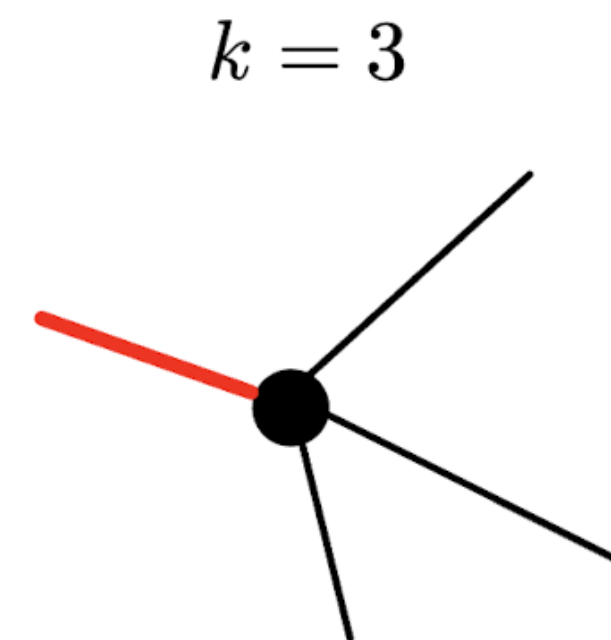
$$q_k = \frac{k+1}{2m} \times np_{k+1} = \frac{(k+1)p_{k+1}}{\langle k \rangle}$$



$$g_1(z) = \frac{g'_0(z)}{g'_0(1)}$$

Excess degree distribution:

$$g_1(z) = \sum_{k=0}^{\infty} q_k z^k$$



Recall $g'_0(1) = \sum_{k=0}^{\infty} kp_k = \langle k \rangle$

Example: Erdos Renyi network:

$$g_0(z) = e^{c(z-1)}$$

$$g_1(z) = e^{c(z-1)}$$

Degree and excess degree have the same distribution!

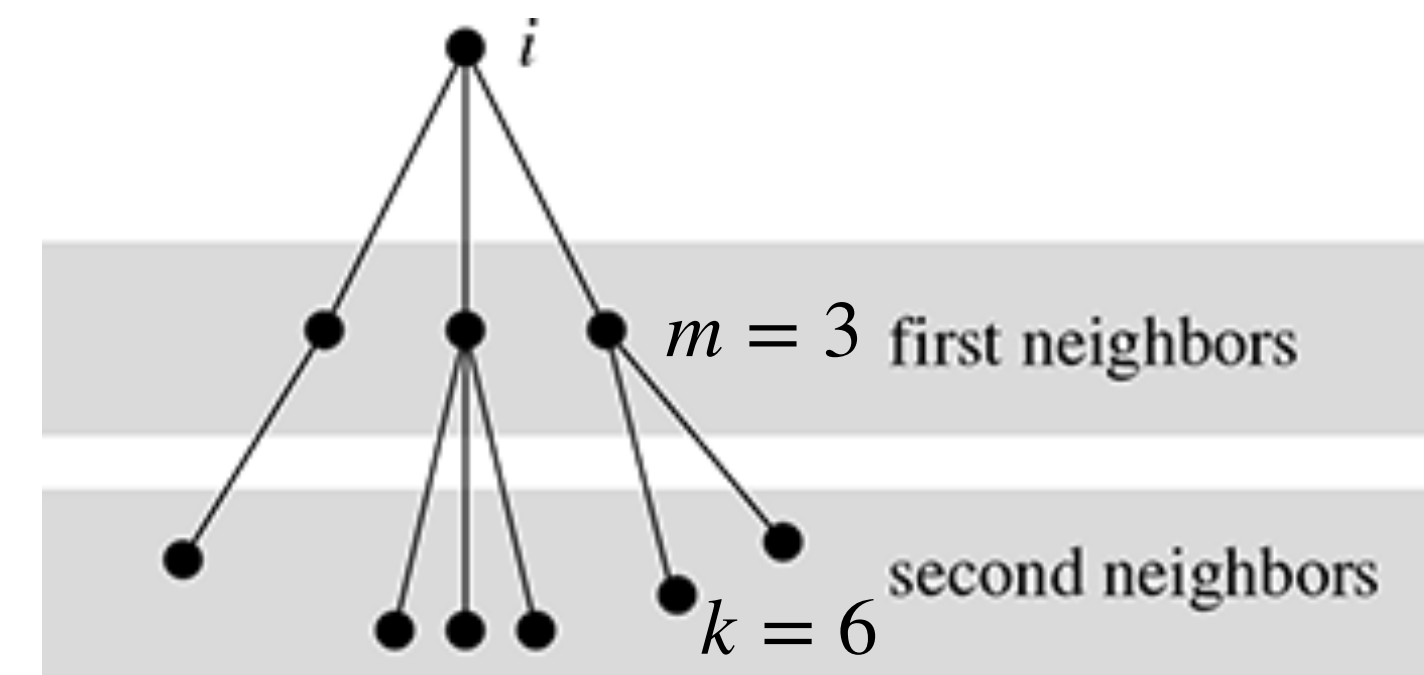
Distribution of 2nd neighbors

Probability distribution of having k 2nd neighbors:

$$p_k^{(2)} = \sum_{m=0}^{\infty} p_m P^{(2)}(k|m) \quad P^{(2)}(k|m) \text{ is the conditional probability that a node with degree } m \text{ has } k \text{ 2nd neighbors}$$

Instead of calculating the probability, we derive its generating function

$$g^{(2)}(z) = \sum_{k=0}^{\infty} p_k^{(2)} z^k = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} p_m P^{(2)}(k|m) z^k = \sum_{m=0}^{\infty} p_m \left[\sum_{k=0}^{\infty} P^{(2)}(k|m) z^k \right]$$



The right most term means the sum of m excess degrees $g_1(z)$. Therefore, by the product property:

$$\sum_{k=0}^{\infty} P^{(2)}(k|m) z^k = [g_1(z)]^m$$

$$g^{(2)}(z) = \sum_{m=0}^{\infty} p_m [g_1(z)]^m$$

Replacing $g_1(z)$ by x (function composition)

$$g^{(2)}(z) = g_0(g_1(z))$$

Average number of 2nd neighbors

Generating function of 2nd neighbors

$$g^{(2)}(z) = g_0(g_1(z))$$

$$g_1(1) = 1 \quad g'_0(1) = \sum_{k=0}^{\infty} k p_k = \langle k \rangle$$

$$\frac{dg^{(2)}}{dz} = g'_0(g_1(z)) g'_1(z)$$

Average 2nd neighbors:

$$c_2 = g'_0(1) g'_1(1)$$

$g'_1(1)$ is average excess degree

$$g'_1(1) = \sum_{k=0}^{\infty} k q_k$$

$$= \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} k(k+1) p_{k+1} = \frac{1}{\langle k \rangle} \sum_{k=0}^{\infty} (k-1) k p_k$$

$$= \frac{1}{\langle k \rangle} (\langle k^2 \rangle - \langle k \rangle),$$

$$q_k = \frac{(k+1)p_{k+1}}{\langle k \rangle}$$

$g'_0(1)$ is average degree

$$g'_0(1) = \langle k \rangle$$

Probability having no 2nd neighbor:

$$p_0^{(2)} = g^{(2)}(0) = g_0(g_1(0)) = g_0(q_0) = g_0(p_1 / \langle k \rangle)$$

New result!

$$c_2 = \langle k \rangle \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = \langle k^2 \rangle - \langle k \rangle$$

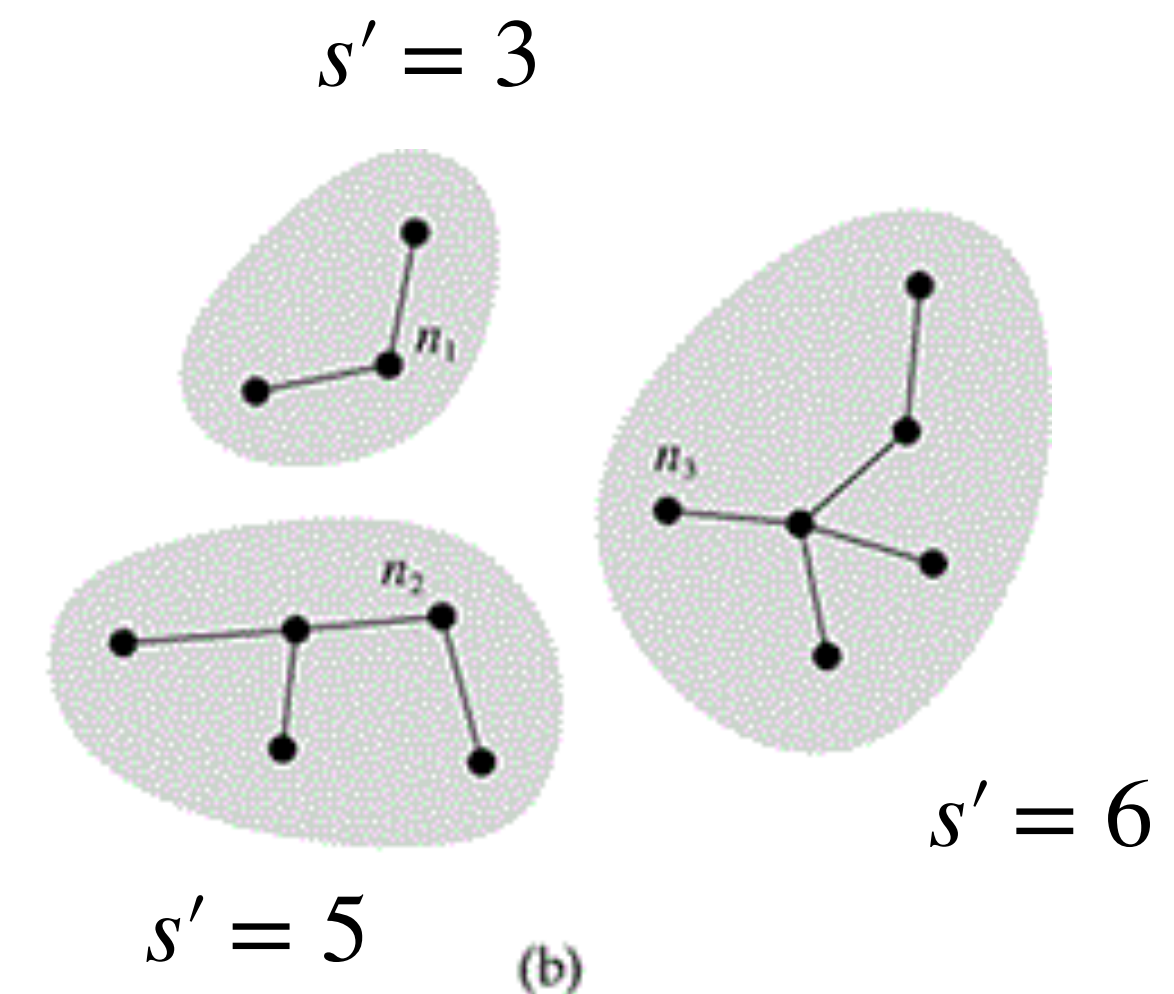
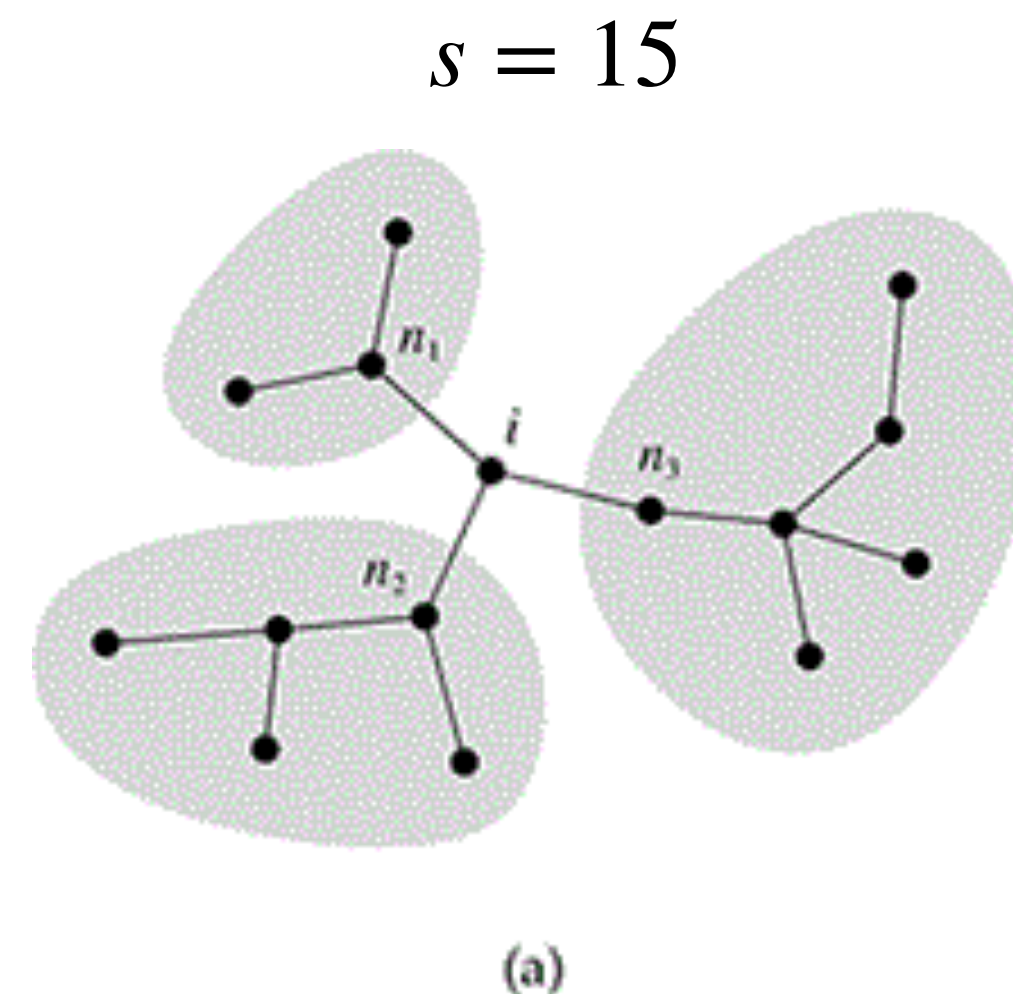
Same as before!

Small components size distribution

π_s

Probability that a randomly chosen node belongs to a small component of size s . Its generating function:

$$h_0(z) = \sum_{s=1}^{\infty} \pi_s z^s$$



$\rho_{s'}$

Probability that a randomly chosen edge leads to sub component size s' . Its generating function:

$$h_1(z) = \sum_{s'=1}^{\infty} \rho_{s'} z^{s'}$$

$h_0(z)$ - focus on node i

$$P(s-1|k)$$

Probability that a node with degree k extends out to $s-1$ other nodes through its k links.

Probability that a randomly chosen node belongs to a small component of size s . Its generating function:

$$\pi_s = \sum_{k=0}^{\infty} p_k P(s-1|k)$$

$$h_0(z) = \sum_{s=1}^{\infty} \pi_s z^s$$

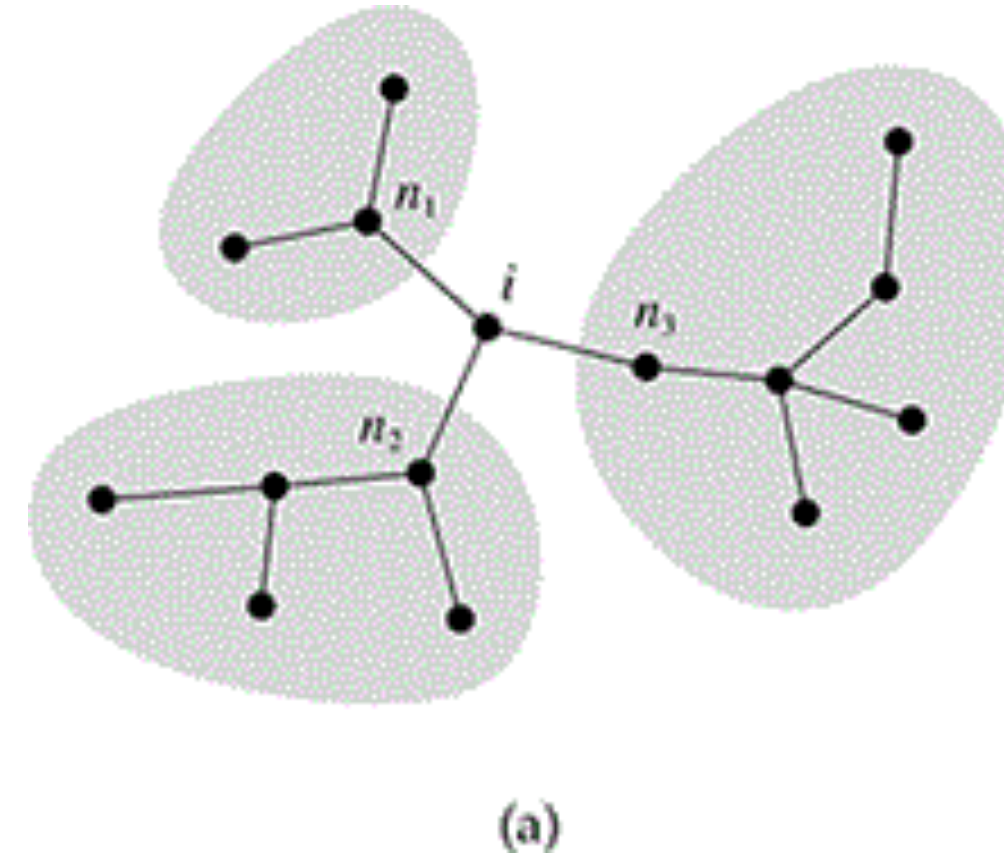
$$\begin{aligned} h_0(z) &= \sum_{s=1}^{\infty} \sum_{k=0}^{\infty} p_k P(s-1|k) z^s = z \sum_{k=0}^{\infty} p_k \sum_{s=1}^{\infty} P(s-1|k) z^{s-1} \\ &= z \sum_{k=0}^{\infty} p_k \sum_{s=0}^{\infty} P(s|k) z^s = z \sum_{k=0}^{\infty} p_k [h_1(z)]^k \end{aligned}$$

$$\sum_s P(s|k) z^s = [h_1(z)]^k$$

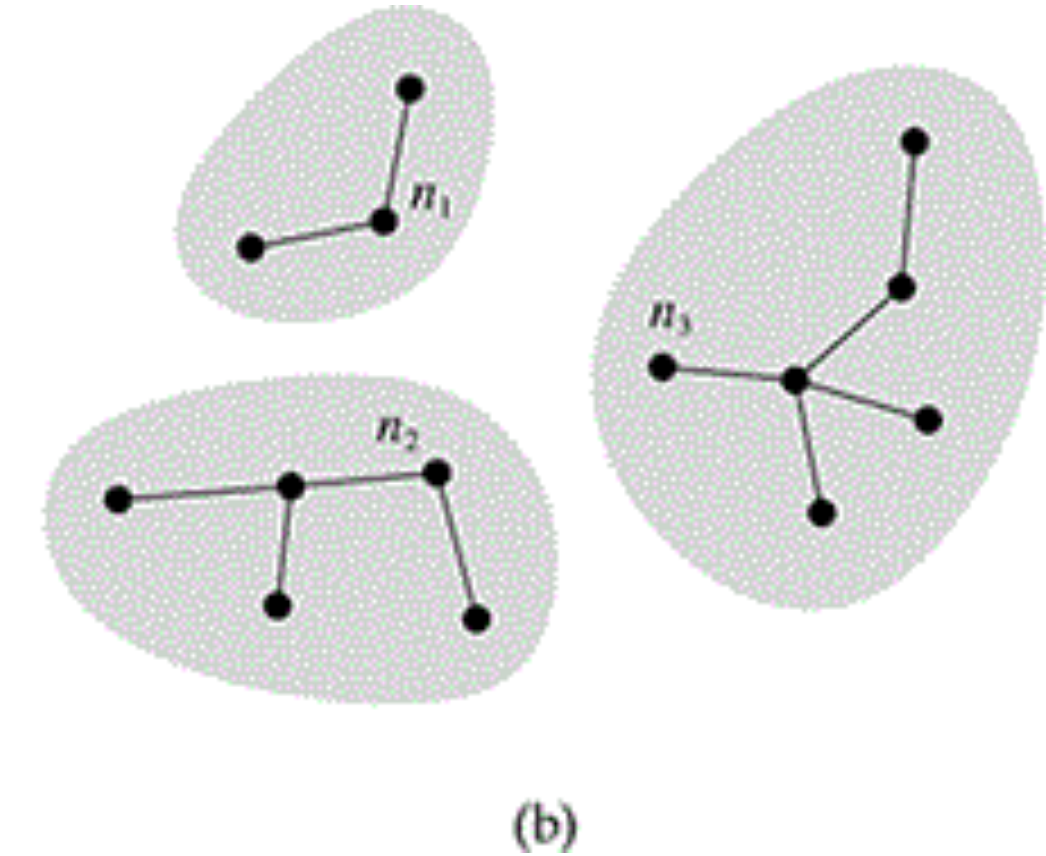
$$h_0(z) = z \sum_{k=0}^{\infty} p_k [h_1(z)]^k = z g_0(h_1(z))$$

Remember $g_0(z)$ is the GF of the network degree distributions

$s = 15$



$s-1 = 14$



$\sum_s P(s|k) z^s$ is the generating function for the **sum** of nodes that can be reached from k links of the node

Apply the product property

$h_1(z)$ - focus on nodes n_i

$P(s - 1 \mid k)$ Probability that a node with excess degree k extends out to $s - 1$ other nodes through its k excess links.

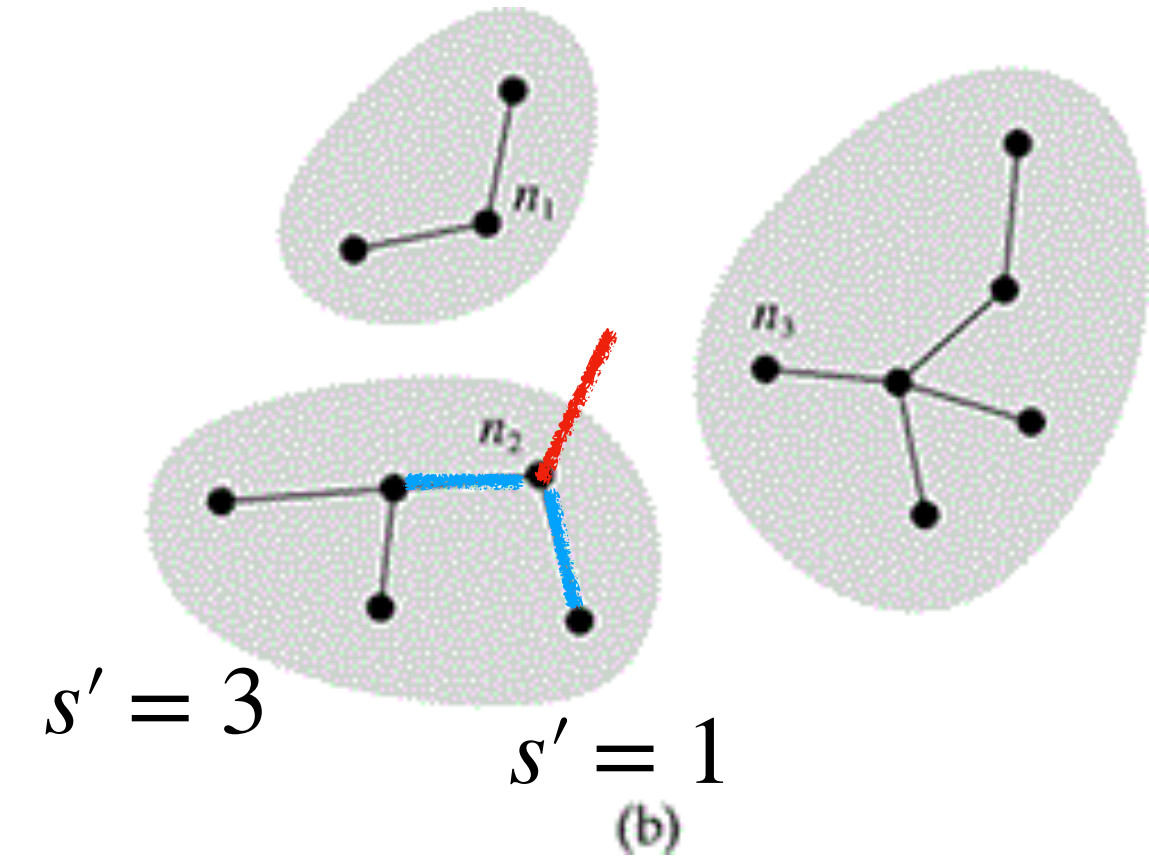
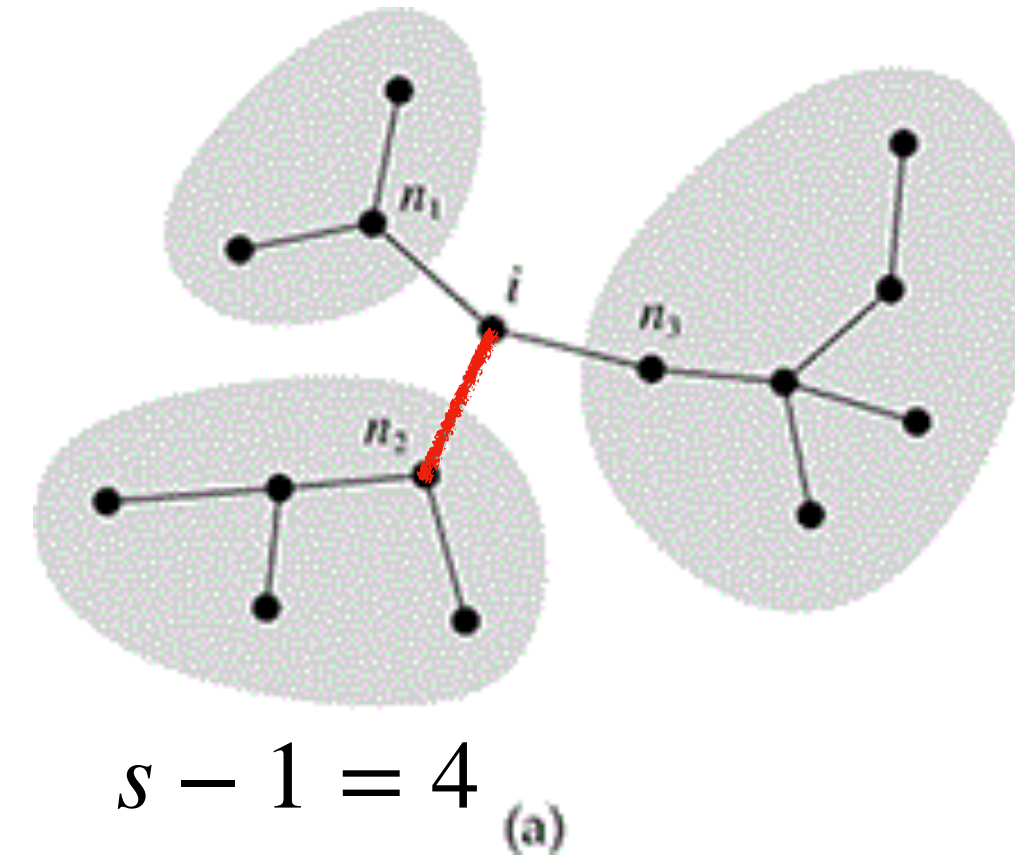
Use excess degree distribution q_k

$$\rho_s = \sum_{k=0}^{\infty} q_k P(s - 1 \mid k)$$

Its generating function by definition is:

$$h_1(z) = \sum_{s=1}^{\infty} \rho_s z^s$$

$$h_1(z) = \sum_{s=1}^{\infty} \sum_{k=0}^{\infty} q_k P(s - 1 \mid k) z^s = z \sum_{k=0}^{\infty} q_k \sum_{s'=0}^{\infty} P(s' \mid k) z^{s'}$$



$\sum_{s'} P(s' \mid k) z^{s'}$ is the generating function for the **sum** of nodes that can be reached from k links of the node

Apply the product property

$$\sum_{s'} P(s' \mid k) z^{s'} = [h_1(z)]^k$$

$$h_1(z) = z \sum_{k=0}^{\infty} q_k [h_1(z)]^k = z g_1(h_1(z))$$

Remember $g_1(z)$ is the GF of the excess degree distributions

Average small components size

When there is no giant component (GC) in the network:

$$h_0(z) = z \sum_{k=0}^{\infty} p_k [h_1(z)]^k = z g_0(h_1(z))$$

Property of GF: mean of distribution is
1st derivative of GF at $z = 1$

$$\begin{aligned} \langle s \rangle &= h'_0(z) |_{z=1} \\ &= \frac{d}{dz} (z g_0(h_1(z))) |_{z=1} \\ &= 1 + g'_0(1) h'_1(1) \end{aligned}$$

$$\begin{aligned} g_0(1) &= 1 \\ h_1(1) &= 1 \end{aligned}$$

$$h_1(z) = z \sum_{k=0}^{\infty} q_k [h_1(z)]^k = z g_1(h_1(z))$$

Differentiate both sides

$$\begin{aligned} h'_1(z) &= 1 + g'_1(1) h'_1(1) \\ &= \frac{1}{1 - g'_1(1)} \end{aligned}$$

$$\begin{aligned} g_1(1) &= 1 \\ h_1(1) &= 1 \end{aligned}$$

$g'_1(1)$ is average excess degree

$$\langle s \rangle = 1 + \frac{g'_0(1)}{1 - g'_1(1)}$$

$$\langle s \rangle \rightarrow \infty, \text{ when } g'_1(1) = 1$$

**Same result as before
on transition point!**

Evaluate the exact distribution

$h_0(z)$ is the GF governing the distribution of the size s of the small component that a randomly selected node i is inside

What are the values of the probabilities π_s inside the GF?

$$h_0(z) = zg_0(h_1(z))$$

$$h_1(z) = zg_1(h_1(z))$$

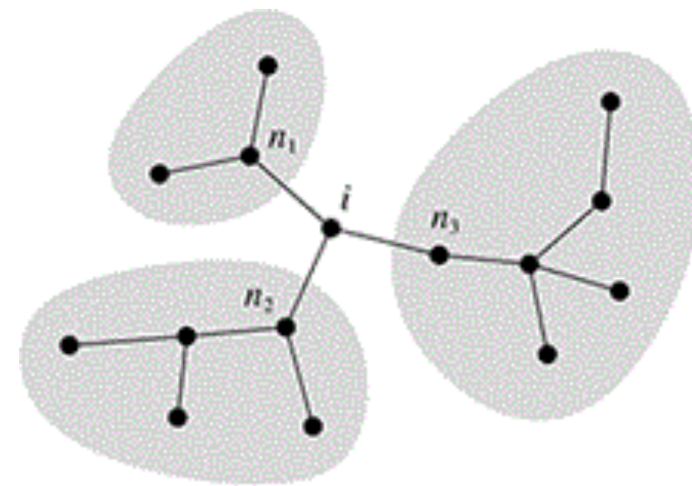
$$h_0(z) = \sum_{s=1}^{\infty} \pi_s z^s$$

Definition of Probability GF

$$\pi_s = \frac{1}{(s-1)!} \left[\frac{d^{s-1}}{dz^{s-1}} \left(\frac{h_0(z)}{z} \right) \right]_{z=0}$$

$$h_0(z) = zg_0(h_1(z))$$

$$\begin{aligned} \pi_s &= \frac{1}{(s-1)!} \left[\frac{d^{s-1}}{dz^{s-1}} g_0(h_1(z)) \right]_{z=0} \\ &= \frac{1}{(s-1)!} \left[\frac{d^{s-2}}{dz^{s-2}} [g'_0(h_1(z)) h'_1(z)] \right]_{z=0} \end{aligned}$$



Cauchy formula at $z_0 = 0$

$$\frac{d^n f}{dz^n} \Big|_{z=z_0} = \frac{n!}{2\pi i} \oint \frac{f(z)}{(z-z_0)^{n+1}} dz$$

$$\pi_s = \frac{1}{2\pi i(s-1)} \oint \frac{g'_0(h_1(z))}{z^{s-1}} \frac{dh_1}{dz} dz$$

$$\pi_s = \frac{1}{2\pi i(s-1)} \oint \frac{g'_0(h_1)}{z^{s-1}} dh_1$$

$$h_1(z) = zg_1(h_1(z))$$

$$\pi_s = \frac{1}{2\pi i(s-1)} \oint \frac{[g_1(h_1)]^{s-1} g'_0(h_1)}{h_1^{s-1}} dh_1$$

$$g_1(u) = \frac{g'_0(u)}{g'_0(1)} \downarrow = \frac{g'_0(1)}{2\pi i(s-1)} \oint \frac{[g_1(h_1)]^s}{h_1^{s-1}} dh_1,$$

Cauchy formula

$$\pi_s = \frac{\langle k \rangle}{(s-1)!} \left[\frac{d^{s-2}}{dz^{s-2}} [g_1(z)]^s \right]_{z=0}$$